Problem 1: State any two conditions under which $AB = BA$, where

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}.$$

Solution

Condition 1: When A and B are diagonal

$$A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & 0 \\ 0 & b_{22} \end{bmatrix},$$

we compute:

$$AB = \begin{bmatrix} a_{11}b_{11} & 0 \\ 0 & a_{22}b_{22} \end{bmatrix}, \quad BA = \begin{bmatrix} b_{11}a_{11} & 0 \\ 0 & b_{22}a_{22} \end{bmatrix}.$$

Since scalar multiplication is commutative, $AB = BA$.

Condition 2: When one matrix is the identity

For $B = I$,

$$AB = A = BA.$$

Similarly, if $A = I$, then $AB = BA = B$.

Problem 2: Convert the matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ into reduced row echelon form.

Solution

Eliminate the entry below the leading 1 by:

$$R_2 \to R_2 - 2R_1 \qquad \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

Eliminate the second-column entry in the first row by:

$$R_1 \to R_1 - 3R_2 \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus, the reduced row echelon form of A is: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Problem 3: If A is a matrix, then $A^r A^s = A^{r+s}$ for positive integers r and s. Does this result hold true for negative integers as well?

Solution

If A is invertible, then for any positive integer n:

$$A^{-n} = (A^n)^{-1}.$$

For negative r, s:

$$A^{-r}A^{-s} = (A^r)^{-1}(A^s)^{-1}.$$

Using the inverse property:

$$(A^s A^r)^{-1} = A^{-(r+s)}.$$

Thus,

$$A^{-r}A^{-s} = A^{-r-s}.$$

Conclusion:
Since $A^{-r}A^{-s} = A^{-(r+s)}$, the property

$$A^r A^s = A^{r+s}$$

holds for negative exponents, **provided A is invertible**.

Problem 4: Show that the matrix

$$A = \begin{bmatrix} 0 & a & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 \\ 0 & d & 0 & e & 0 \\ 0 & 0 & f & 0 & g \\ 0 & 0 & 0 & h & 0 \end{bmatrix}$$

is not invertible for any values of its entries.

Solution

We augment A with the identity matrix:

$$\left[\begin{array}{ccccc|ccccc} 0 & a & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & d & 0 & e & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & f & 0 & g & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

To make leading entries of each row 1 we apply following row operations assuming a, b, d, f and h are non-zero.

$$\left[\begin{array}{ccccc|ccccc} 0 & 1 & 0 & 0 & 0 & \frac{1}{a} & 0 & 0 & 0 & 0 \\ 1 & 0 & \frac{c}{b} & 0 & 0 & 0 & \frac{1}{b} & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{e}{d} & 0 & 0 & 0 & \frac{1}{d} & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{g}{f} & 0 & 0 & 0 & \frac{1}{f} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{h} \end{array}\right] \qquad \begin{aligned} R_1 &\to \frac{1}{a}R_1 \\ R_2 &\to \frac{1}{b}R_2 \\ R_3 &\to \frac{1}{d}R_3 \\ R_4 &\to \frac{1}{f}R_4 \\ R_5 &\to \frac{1}{h}R_5 \end{aligned}$$

Since $a_{11} = 0$, we interchange R_1 and R_2:

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & \frac{c}{b} & 0 & 0 & 0 & \frac{1}{b} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{a} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{e}{d} & 0 & 0 & 0 & \frac{1}{d} & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{g}{f} & 0 & 0 & 0 & \frac{1}{f} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{h} \end{array}\right] \qquad R_1 \leftrightarrow R_2$$

Eliminate second column below leading 1,

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & \frac{c}{b} & 0 & 0 & 0 & \frac{1}{b} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{a} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{e}{d} & 0 & \frac{-1}{a} & 0 & \frac{1}{d} & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{g}{f} & 0 & 0 & 0 & \frac{1}{f} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{h} \end{array}\right] \qquad R_3 \to R_3 - R_2$$

Observe that R_3 proportional to R_5, so we can use R_5 to eliminate R_3 completely.

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & \frac{c}{b} & 0 & 0 & 0 & \frac{1}{b} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{a} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{-1}{a} & 0 & \frac{1}{d} & 0 & \frac{-e}{dh} \\ 0 & 0 & 1 & 0 & \frac{g}{f} & 0 & 0 & 0 & \frac{1}{f} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{h} \end{array}\right] \qquad R_3 \to R_3 - \frac{e}{d}R_5$$

This means that A can not have it's reduced row echelon form as Identity and hence has no inverse.

Problem 5: Convert the matrix $A = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$ into reduced row echelon form.

Solution

We reduce A to its reduced row echelon form.
Interchange rows by:

$$R_1 \leftrightarrow R_2 \qquad \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$$

Eliminate the entry below the leading 1 by:

$$R_2 \to R_2 - 2R_1 \qquad \begin{bmatrix} 1 & 5 \\ 0 & -6 \end{bmatrix}$$

Normalize the pivot at $(2, 2)$ by:

$$R_2 \to \frac{R_2}{-6} \qquad \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

Eliminate the second-column entry in the first row by:

$$R_1 \to R_1 - 5R_2 \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus, the reduced row echelon form of A is: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Problem 6: Show that:

$$(A^{-1})^T = (A^T)^{-1}.$$

Solution

We start with:

$$AA^{-1} = I.$$

Taking the transpose:

$$(AA^{-1})^T = I^T.$$

Since $I^T = I$, we get:

$$(A^{-1})^T A^T = I.$$

This shows A^T and $(A^{-1})^T$ satisfy the inverse property:

$$(A^{-1})^T A^T = I.$$

Thus, by definition,

$$(A^{-1})^T = (A^T)^{-1}.$$

Problem 7: Solve the following system of linear equations using the **Gaussian elimination** method:

$$\begin{aligned} x + y + 2z &= 9, \\ 2x + 4y - 3z &= 1, \\ 3x + 6y - 5z &= 0. \end{aligned}$$

Solution

We represent the system as an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right].$$

Eliminate the first-column entries below the pivot by:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \qquad \begin{aligned} R_2 &\to R_2 - 2R_1 \\ R_3 &\to R_3 - 3R_1 \end{aligned}$$

Normalize the second pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right] \qquad R_2 \to \frac{R_2}{2}$$

Eliminate the second-column entry in R_3:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & \frac{-1}{2} & \frac{-3}{2} \end{array}\right] \qquad R_3 \to R_3 - 3R_2$$

Normalize the third pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right] \qquad R_3 \to -2R_3$$

Back-substitution:
From R_3: $z = 3$.
From R_2:

$$y - \frac{7}{2}(3) = -\frac{-17}{2} \quad \Rightarrow \quad y - \frac{21}{2} = -\frac{17}{2} \quad \Rightarrow \quad y = 2.$$

From R_1:

$$x + 2 + 6 = 9 \quad \Rightarrow \quad x = 1.$$

Final Answer:

$$\boxed{x = 1, \quad y = 2, \quad z = 3.}$$

Problem 8: Solve the following system of linear equations using the **Gauss-Jordan elimination** method:

$$\begin{aligned} 2x + 2y + 2z &= 0, \\ -2x + 5y + 2z &= 1, \\ 8x + y + 4z &= -1. \end{aligned}$$

Solution

We represent the system as an augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right].$$

Normalize the first pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \qquad R_1 \to \tfrac{R_1}{2}$$

Eliminate the first-column entries below the pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \qquad \begin{aligned} R_2 &\to R_2 + 2R_1 \\ R_3 &\to R_3 - 8R_1 \end{aligned}$$

Normalize the second pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & -7 & -4 & -1 \end{array}\right] \qquad R_2 \to \frac{R_2}{7}$$

Eliminate second-column entries:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad \begin{aligned} R_1 &\to R_1 - R_2 \\ R_3 &\to R_3 + 7R_2 \end{aligned}$$

The third row is all zeros, indicating that z is a free variable. Let $z = t$ then:

$$y + \frac{4}{7}t = \frac{1}{7} \quad \Rightarrow \quad y = \frac{1}{7} - \frac{4}{7}t.$$

$$x + \frac{3}{7}t = -\frac{1}{7} \quad \Rightarrow \quad x = -\frac{1}{7} - \frac{3}{7}t.$$

Final Answer (Parametric Form):

$$\boxed{x = -\frac{1}{7} - \frac{3}{7}t, \quad y = \frac{1}{7} - \frac{4}{7}t, \quad z = t, \quad t \in \mathbb{R}.}$$

Problem 9: For what values of a does the system of equations have "no solution" and "a unique solution"?

$$ax + y = 3,$$
$$x + ay = 5.$$

Solution

We represent the system as an augmented matrix:

$$\left[\begin{array}{cc|c} a & 1 & 3 \\ 1 & a & 5 \end{array}\right].$$

Eliminate the first-column entry in R_2:

$$\left[\begin{array}{cc|c} a & 1 & 3 \\ 0 & a - \frac{1}{a} & 5 - \frac{3}{a} \end{array}\right] \qquad R_2 \to R_2 - \frac{1}{a}R_1$$

For a unique solution, the second pivot must be nonzero:

$$a - \frac{1}{a} \neq 0.$$

$$a^2 \neq 1.$$

$$a \neq \pm 1.$$

Now checking for $a = \pm 1$:
If $a = 1$ second row becomes:

$$0x + 0y = 5 - 3 = 2 \quad (0 = 2) \quad \Rightarrow \quad \text{No Solution.}$$

If $a = -1$ second row becomes:

$$0x + 0y = 5 + 3 = 8 \quad (0 = 8) \quad \Rightarrow \quad \text{No Solution.}$$

Conclusion:

Unique solution for $a \neq \pm 1$.
No solution for $a = \pm 1$.

Problem 10: For what values of b does the system of equations have "a unique solution" and "no solution"?

$$bx + y = 1,$$
$$x + by = 1.$$

Solution

We represent the system as an augmented matrix:

$$\left[\begin{array}{cc|c} b & 1 & 1 \\ 1 & b & 1 \end{array}\right].$$

Eliminate the first-column entry in R_2:

$$\left[\begin{array}{cc|c} b & 1 & 1 \\ 0 & b - \frac{1}{b} & 1 - \frac{1}{b} \end{array}\right] \qquad R_2 \to R_2 - \frac{1}{b} R_1$$

For a unique solution, the second pivot must be nonzero:

$$b - \frac{1}{b} \neq 0.$$

$$b^2 \neq 1.$$

$$b \neq \pm 1.$$

Now checking for $b = \pm 1$:
If $b = 1$ the second row becomes:

$$0x + 0y = 1 - 1 = 0 \quad \text{(infinitely many solutions)}.$$

If $b = -1$ the second row becomes:

$$0x + 0y = 1 + 1 = 2 \quad (0 = 2) \quad \Rightarrow \quad \text{No Solution}.$$

Conclusion:

Unique solution for $b \neq \pm 1$.
No solution for $b = -1$.
Infinitely many solutions for $b = 1$.

Problem 11: f A is an invertible matrix such that

$$A^3 = A^2 + A - 5I,$$

prove that

$$A^{-1} = \frac{1}{5}(I + A - A^2).$$

Solution

Rearrange the given equation:

$$A^3 - A^2 - A + 5I = 0.$$

Multiply by A^{-1}:

$$A^{-1}(A^3 - A^2 - A + 5I) = 0.$$

Expanding:

$$A^{-1}A^3 - A^{-1}A^2 - A^{-1}A + 5A^{-1} = 0.$$

Using $A^{-1}A = I$, $A^{-1}A^2 = A$, and $A^{-1}A^3 = A^2$:

$$A^2 - A - I + 5A^{-1} = 0.$$

Solving for A^{-1}:

$$5A^{-1} = I + A - A^2.$$

$$A^{-1} = \frac{1}{5}(I + A - A^2).$$